Agricultural Adaptation to Drought: Evidence from Vietnam*

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Abstract

We study how drought shapes agricultural adaptation through the reallocation of land and inputs in Vietnam. Using a long difference approach, we find that in the long run, drought reallocates land away from rice toward other crops, while the shift toward aquaculture depends on local water conditions. Our mechanism analysis shows that drought persistently reduces rice productivity. Irrigation infrastructure mitigates the contraction of rice and facilitates crop diversification, while electricity access enables mechanization in aquaculture. We also find that drought is associated with a shift of labor from manufacturing into agriculture, where it is absorbed primarily by crop cultivation rather than aquaculture. Our findings shed light on how infrastructure and local factor markets shape farmers' long-run adaptation margins.

Keywords: drought, adaptation, agricultre, land use

JEL Codes: Q54, Q15, Q12, O13

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1 Introduction

Drought is considered one of the most damaging climate shocks for global agriculture and food security (Zhao and Running, 2010; Lesk et al., 2016). In developing countries, where markets are incomplete and coping mechanisms are limited, farming households that depend on land productivity are particularly vulnerable. An important margin of adaptation is therefore how agricultural land is reallocated across activities with different exposure to water scarcity and different input requirements. Understanding these land-use responses, and the local conditions that enable or constrain them, is important for designing policies that strengthen agricultural resilience to climate change (Howden et al., 2007; Deressa et al., 2009).

Land reallocation is not necessarily a frictionless response to climate stress. Alternative agricultural activities differ in their dependence on water, electricity, and other productive inputs, so the direction of adaptation may depend on local resource endowments and infrastructure. Rice cultivation is often sensitive to freshwater availability, while intensive aquaculture may depend relatively more on electricity powered machinery. These differences suggest that local infrastructure can shape how agricultural production adjusts to climate shocks. Irrigation infrastructure can buffer production against rainfall shocks (Duflo and Pande, 2007; Hornbeck and Keskin, 2014), whereas reliable electricity access may facilitate adjustment toward more intensive aquaculture. Despite growing evidence that climate change affects crop choice and agricultural acreage (Burke and Emerick, 2016; Cui, 2020b), less is known about how resource constraints and infrastructure shape land use adaptation, particularly in developing countries.

This paper examines how drought reshapes agricultural land use in Vietnam and how local conditions influence the direction of adaptation. We address two main questions. First, how does drought reallocate land among rice paddies, other crops, and aquaculture in the short and long run? Second, how do local differences in water conditions and infrastructure, especially irrigation and electricity access, shape these adjustments? Vietnam provides a particularly useful setting for addressing these questions. As a major rice producer and exporter, the country is highly exposed to drought and salinity risks, while agricultural activities differ substantially in their dependence on freshwater and electricity. We leverage these differences to examine not only whether drought shifts land away from rice, but also which alternative uses emerge and which local constraints shape those transitions. We further provide complementary evidence on how drought-induced disruptions outside agriculture, particularly in electricity-intensive manufacturing, are associated with machinery and labor adjustments within agriculture.

We first analyze the short-run and long-run effects of drought on agricultural land use using nationally comprehensive satellite data from 2000 to 2020, aggregated to the commune level. These data allow us to track land allocated to rice paddies, other crops, and aquaculture, together with drought exposure and local water conditions. We complement the land-use data with administrative and remote-sensing measures of rice productivity, as well as information on irrigation and electricity infrastructure, to examine the mechanisms underlying land reallocation. Agricultural Census and Enterprise Census data provide additional evidence on machinery use, agricultural labor, and manufacturing activity. Our preferred approach for studying long-run adaptation is a long-difference design, following Burke and Emerick (2016); Cui and Zhong (2024), which compares changes in average drought exposure and land use between 2000–2005 and 2015–2020. We compare these long-run estimates with those from a conventional panel fixed-effects specification that exploits annual within-commune variation in drought exposure, following (Deschênes and Greenstone, 2007; Schlenker and Roberts, 2009; Cui, 2020b). The panel estimates capture shorter-run responses to annual drought shocks, while the long-difference estimates capture more persistent land-use adjustments associated with sustained changes in drought exposure.

We report three main findings on agricultural land adaptation and additional evidence on broader intersectoral adaptation. First, drought systematically reallocates agricultural land away from rice. In the short run, a one-standard-deviation increase in drought reduces rice paddy area by about 4.6% of its baseline mean, while increasing other cropland by 2.8% and aquaculture by 25.6%. In the long run, rice area remains significantly lower and the expansion of other cropland becomes larger, consistent with persistent crop diversification. The average long-run aquaculture effect is positive but imprecisely estimated, suggesting that aquaculture adjustment is more dependent on local conditions. These results show that adaptation occurs not only through changes in agricultural productivity, but also through changes in the composition of agricultural land.

Second, the direction of land-use adaptation depends importantly on local water conditions. Drought produces larger declines in rice area in locations with greater baseline salinity, while the aquaculture response is nonmonotonic in that aquaculture expands most clearly at intermediate salinity levels but contracts where salinity is very high. These patterns suggest that drought does not generate a uniform shift from rice into a single alternative activity. Instead, the feasible adaptation path depends on local ecological conditions, particularly the quantity and quality of available water.

Third, we identify mechanisms that help explain these land-use transitions. Greater

drought exposure persistently reduces rice yields, lowering the returns to maintaining land in rice. At the same time, the ability to adjust depends on infrastructure. Areas with greater irrigation infrastructure experience smaller drought-induced reductions in rice area and stronger expansion of other crops, indicating that water infrastructure can both protect existing production and facilitate diversification. Electricity access plays a different role: aquaculture expands substantially more in areas with better access to the transmission network, consistent with the sector's greater reliance on electricity-intensive machinery. Together, these results show that land-use adaptation reflects both changes in relative returns and differences in the feasibility of alternative agricultural activities.

As additional evidence on broader adjustment to drought and the dynamic interaction between agriculture and non-agriculture sectors, we examine responses in manufacturing industries, specifically in labor reallocation and agricultural mechanization. Drought substantially reduces manufacturing activity, with larger effects on electricity-intensive industries, consistent with drought-induced pressure on electricity supply. Labor adjusts across sectors accordingly. Non-agricultural employment falls, mainly in manufacturing, while crop-cultivation employment rises. Aquaculture employment does not increase despite expansion in aquaculture-related activities. Instead, the use of aquaculture machineries expand, primarily in districts with a lower presence of electricity-intensive manufacturing firms. These results provide complementary evidence that the land-use responses we document are embedded in broader adjustments in local input and factor markets.

First, our paper extends the understanding of how drought and climate stress affect agriculture in developing countries, particularly through changes in land allocation. A large literature documents substantial effects of heat, drought, and water stress on crop yields and agricultural production (Schlenker and Roberts, 2009; Kuwayama et al., 2019; Cui, 2020a; Aragón et al., 2021; Wang et al., 2022). Much of the evidence on land-use and acreage adjustments to climate shocks, however, comes from developed agricultural economies, where farmers respond by reallocating acreage across major crops (Cui, 2020b; Sumner et al., 2026). A growing literature also examines the effects of temperature and precipitation shocks in developing countries (Welch et al., 2010; Nguyen and Scrimgeour, 2022; Chen et al., 2023; Chen and Gong, 2021). Some recent studies document land-use responses, including cropland expansion following weather shocks in Ethiopia and climate-induced crop substitution in Vietnam (He and Chen, 2022; Nguyen Chau and Scrimgeour, 2023). Nevertheless, relatively little evidence documents how farmers in developing countries respond to drought by persistently reallocating land across

competing agricultural uses (Salazar-Espinoza et al., 2015; Tabe-Ojong et al., 2026). We fill this gap by studying Vietnam, a major rice-producing economy that is highly exposed to drought and vulnarable to climate stress. Rather than focusing only on yield or production effects, we show that drought induces persistent changes in land allocation, reducing land devoted to rice while increasing land allocated to other crops, with additional adjustments toward aquaculture depending on local conditions. Our findings therefore identify land reallocation as an important margin of drought agriculutral adaptation to drought in in developing countries.

Second, we show how local infrastructure shapes heterogeneous agricultural responses to drought when natural water availability is limited and agricultural activities differ in their input requirements. This setting is particularly relevant in Vietnam, where 99% of the area equipped for irrigation relies on surface water and only 1% on groundwater (Frenken, 2012). A growing literature shows that irrigation systems, dams, and groundwater access can sustain agricultural production and mitigate the effects of drought and other climatic shocks (Duflo and Pande, 2007; Olmstead and Sigman, 2026; Taylor, 2026; Cisse et al., 2026). We extend this literature by showing that irrigation infrastructure also shapes land reallocation across agricultural uses. Where irrigation water is more readily available, drought leads to smaller contractions in freshwater intensive rice cultivation and greater expansion of other crops. We further show that electricity infrastructure shapes adjustment toward activities with greater electricity requirements. Existing work highlights the importance of electricity for agricultural production through its complementarity with irrigation and groundwater pumping, and shows that electrification can affect productivity and land allocation across agricultural activities (Fishman et al., 2023; Szerman et al., 2026). Given the limited role of groundwater in Vietnamese irrigation, we instead find that better electricity access facilitates reallocation toward electricity intensive aquaculture.[1] Our findings therefore highlight a broader role for infrastructure in climate adaptation. Different forms of infrastructure relax different input constraints and thereby shape the direction of land reallocation. Irrigation infrastructure helps sustain freshwater dependent production and supports adjustment across crops, while electricity infrastructure enables adjustment toward electricity intensive aquaculture.

Third, our study contributes to a large literature documenting labor reallocation in response to climate driven changes in agricultural productivity (Colmer, 2021; Blakeslee

[1]Rice is particularly water intensive, with total water inputs to lowland rice fields reaching two to three times those of other major cereals (Steduto, P. et al., 2012). Aquaculture is also energy dependent, requiring electricity to power equipment such as pumps and aerators (FAO, 2022).

et al., 2020; Grabrucker and Grimm, 2021). A smaller literature examines how local labor market conditions shape these responses (Liu et al., 2023; Pham, 2025).[2] Liu et al. (2023) emphasize local labor market adjustment as an important mechanism of climate driven labor reallocation, while Pham (2025) show that local conditions, particularly trade openness, generate heterogeneity in these responses. Albert et al. (2021) further show that limited capital reallocation can constrain structural adjustment, with workers moving primarily toward small firms outside manufacturing.

We extend this literature by showing that climate shocks affect not only workers' incentives to leave agriculture but also labor demand in the sectors into which they might otherwise reallocate. This intersectoral channel is particularly important in Vietnam, where reliance on hydropower means that drought can simultaneously reduce agricultural output and disrupt electricity intensive production. Hydropower accounted for roughly 55% of Vietnam's electricity generation in 2000, and drought driven electricity shortages have exposed firms to hydrological shocks (OECD, 2021; Lee and Gerner, 2020; Elliott et al., 2021). Related evidence shows that electricity shortages associated with hydroelectricity can adversely affect manufacturing plants (Allcott et al., 2016). Using direct measures of local labor demand outside agriculture, particularly in manufacturing, we show that drought weakens employment opportunities outside agriculture and induces workers to reallocate back toward agriculture. More broadly, our results show that climate damages can propagate across sectors through shared infrastructure rather than remain confined to the sector directly exposed to the shock. Finally, we document the labor and capital dynamics underlying adaptation within agriculture. Rice intensification and crop diversification absorb additional agricultural labor, whereas aquaculture expansion is driven primarily by machinery adoption rather than labor inflows.

The rest of the paper proceeds as follows. Section 2 describes the data used in our analysis, ranging from census records to satellite imagery. Section 3 presents the empirical strategy. Section 4 reports reports the main results, examines heterogeneous effects of drought on land use, and presents a series of robustness checks. Section 5 examines the underlying mechanisms, while Section 6 presents additional evidence on intersectoral adaptation. Section 7 concludes.

[2]Relatedly, Afridi et al. (2022) find that social norms may constitute an additional labor market friction that limits women's transitions into nonfarm employment.

2 Data

We construct a nationally comprehensive commune-by-year panel dataset for Vietnam by integrating data on land use, drought, hydrology, infrastructure, labor, and agricultural machinery from multiple sources over the period 2000–2020. We use this dataset to examine the effects of drought on land use change and the underlying mechanisms.

Drought. We measure drought exposure using the Palmer Drought Severity Index (PDSI) (Palmer, 1965) from TerraClimate (Abatzoglou et al., 2018). TerraClimate provides monthly climate and PDSI measures, which we overlay with commune boundaries and aggregate to the commune-year level.[3] We focus on drought conditions by setting non-negative PDSI values to zero and reversing the sign of negative values, such that larger values indicate greater drought intensity.[4]

Land Use. We study agricultural land use diversification and structural change, focusing on the reallocation of land among rice paddies, other crops, and aquaculture. To measure these outcomes, we use JAXA's remotely sensed Vietnam Land-Use/Land-Cover Data Sets (VLUCDs), which provide annual land-use and land-cover classifications across mainland Vietnam from 2000 to 2020 (Duong et al., 2018; Phan et al., 2021, 2022). We overlay the annual land use classifications with commune boundaries to calculate the area allocated to each of the three categories for every commune and year, measured in hectares (ha).[5]

Water. Surface water is the predominant source of irrigation water in Vietnam. According to the FAO, 99% of the area equipped for irrigation relied on surface water in 2005, compared with only 1% relying on groundwater (Frenken, 2012). We therefore focus on surface water availability and quality as baseline hydrological conditions. We measure surface water availability by calcualting permanent surface water area within each commune using the Global Surface Water Explorer data (Pekel et al., 2016). We measure surface water quality using salinity concentrations (mg/L) from Jones et al.

[3] TerraClimate is a global monthly climate dataset available from 1958 onward. In addition to temperature and precipitation, it provides hydro-climate variables derived from a modified Thornthwaite–Mather water-balance model and reference evapotranspiration based on the ASCE Penman–Monteith method, including soil moisture, runoff, climatic water deficit, and PDSI (Abatzoglou et al., 2018).

[4] We follow the approach commonly used in the literature to construct a drought-focused measure from PDSI (Hornbeck and Keskin, 2014; Sumner et al., 2026; Won et al., 2024).

[5] We use the VLUCD Level-1 classification with ten primary land use classes. VLUCD provides annual 30 m classifications constructed from Landsat and Sentinel-1/-2 imagery using random-forest classification, post-classification optimization, and independently validated reference data. The reported overall accuracy is approximately 86–92% for the Level-1 classification and 78–85% for the Level-2 classification (Phan et al., 2021). The other crops category combines woody crops and in-house crops, including coffee, tea, orchards, and crops cultivated in greenhouses or other protected structures. The ten Level-1 land use categories are summarized in Table A1.

(2023). We overlay both raster datasets with commune boundaries to construct baseline measures of local surface water conditions.

Infrastructure. We compile data on water and electricity infrastructure to study whether existing water infrastructure mitigates the effects of drought on land use. We measure baseline water infrastructure using the number of irrigation dams. Dam locations are obtained from administrative records from Vietnam's Department of Irrigation.[6] For each commune, we measure local dam infrastructure using the number of irrigation dams in its district and upstream dam infrastructure using the number of dams in neighboring upstream districts. To identify upstream districts, we link district boundaries to HydroBASINS and trace the river network upstream to the first hydrologically connected neighboring district, following the spatial logic of Duflo and Pande (2007) and Strobl and Strobl (2011). We also compile data on electricity infrastructure. Our main measure of electricity infrastructure access is proximity to transmission lines. We obtain georeferenced electricity transmission networks for 2005 and 2016 from the World Bank's ENERGYDATA.INFO platform. We calculate each commune's distance to the nearest transmission line as a proxy for access to electricity infrastructure.[7]

Machinery and Labor. We use Vietnam's Rural, Agricultural, and Fisheries Census (RAFC), hereafter the Agricultural Census, to construct measures of agricultural machinery use and labor.[8] We use the number of water pumps per household as a proxy for demand for supplemental irrigation and the number of generators per household as a proxy for demand for supplemental electricity. We also construct measures of rice threshers, aquaculture aerators, and processing machinery to examine machinery-based adaptation. For labor, we aggregate household-level employment in primary agricultural and non-agricultural activities to construct commune-level counts of workers in each sector. We complement these measures with district-year data on manufacturing employment and the number of manufacturing firms, defined using two-digit industry codes from the annual Vietnam Enterprise Census (VEC) for 2006–2016.

[6]Dam records are from the Department of Irrigation, Ministry of Agriculture and Rural Development, *Dam Technical Parameters Report*, https://baocaonhanh.thuyloivietnam.vn/ (report generated August 21, 2026). The records cover 1,957 weirs across 23 provinces. We assign each dam to a district by matching its reported location to commune and district names.

[7]The datasets are available at https://energydata.info/dataset/vietnam-electricity-transmission-network (2005) and https://energydata.info/dataset/vietnam-electricity-transmission-network-2016 (2016).

[8]The Agricultural Census is nationally representative and was administered in 2000, 2006, 2011, and 2016. It enumerates rural households and urban households engaged in agriculture, forestry, or fisheries across all 63 provinces. Each round includes a universal household questionnaire covering household characteristics, labor, land use, livestock, and machinery and equipment. We aggregate variables from the universal household questionnaire to the commune level to construct a panel spanning 2000–2016.

Other Data. We use province-level paddy yields from the National Statistics Office of Vietnam (NSO).[9] For cross-validation, we compare these statistics with remote-sensing-based rice yield estimates from Wu et al. (2023). We also use population density from WorldPop (Tatem, 2017), nighttime lights (Chen et al., 2021), and distance to the coast (NASA Ocean Biology Processing Group (OBPG) and Stumpf, R. P., 2012) as baseline controls for potential confounding factors. Finally, we use rice cropping-calendar data from RiceAtlas (Laborte et al., 2017) to align the timing of rice cultivation with drought exposure.

Summary Statistics. Across the 11,027 communes in our sample, the long-run average drought intensity, calculated using the two periods 2000–2005 and 2015–2020, is 1.57 PDSI units, with a standard deviation of 1.07. In 2000, the average commune had 421.6 hectares of rice paddies, 507.7 hectares of other cropland, and 26.8 hectares of aquaculture, as reported in Table 1. Nationally, drought conditions fluctuate considerably over the study period rather than following a monotonic trend. Drought intensity rises sharply in the early 2000s and reaches a pronounced peak around 2005, with additional dry episodes around 2010 and 2015. Conditions become less dry in the late 2010s before drought intensity increases again by 2020. Rice and other cropland exhibit opposing trends. Rice area increases through the late 2000s as other cropland declines, while after 2009 rice area declines as other cropland expands. Aquaculture increases steadily throughout the sample period, as shown in Figure 1.

These national trends mask substantial spatial heterogeneity. Drought exposure increases across much of central Vietnam and the northwest but declines in parts of the northeast, southeast, and Mekong Delta, as shown in Figure 2. Rice area expands mainly in the north but contracts across parts of the major deltas, while other cropland expands primarily in central Vietnam. Aquaculture expansion is concentrated in the Mekong Delta and other coastal areas, as shown in Figure 3.

3 Empirical Strategy

We use a long-difference approach to identify the long-run effects of drought on land use, following Burke and Emerick (2016)[10]. We define the starting period as 2000–2005 and the ending period as 2015–2020. Averaging outcomes over multiple years at the

[9] Paddy yields are from the NSO's *Yield of Paddy by Province* series, available through the NSO PX-Web database at `https://www.nso.gov.vn/en`.

[10] This long-difference apparoach has also been used to study adaptation in crop production by Chen and Gong (2021); Cui and Zhong (2024); Schmitt et al. (2022), etc.

beginning and end of the sample reduces the influence of short-run fluctuations and allows us to examine whether persistent changes in drought conditions are associated with long-run changes in land use.

For any variable Z_{irt}, where i indexes communes, r indexes regions, and t indexes years, we calculate the long difference as the change in its average value between the ending and starting periods:

$$\Delta \overline{Z}_{ir} = \overline{Z}_{ir}^{\text{end}} - \overline{Z}_{ir}^{\text{start}} = \frac{1}{N_{\text{end}}} \sum_{t \in T^{\text{end}}} Z_{irt} - \frac{1}{N_{\text{start}}} \sum_{t \in T^{\text{start}}} Z_{irt}, \tag{1}$$

where T^{start} and T^{end} denote the sets of years in the starting and ending periods, respectively, and N_{start} and N_{end} denote the corresponding numbers of years. Using these long differences, we estimate the following regression model:

$$\Delta \overline{Y}_{ir} = \beta^{LD} \Delta \overline{D}_{ir} + \delta_r + \Delta \varepsilon_{ir}, \tag{2}$$

where $\Delta \overline{Y}_{ir}$ denotes the long-run change in the land-use outcome of commune i in region r, including the area of rice paddies, other cropland, or aquaculture. Specifically, $\Delta \overline{Y}_{ir} = \left(\frac{1}{6}\sum_{t=2015}^{2020} Y_{irt}\right) - \left(\frac{1}{6}\sum_{t=2000}^{2005} Y_{irt}\right)$. The key independent variable, $\Delta \overline{D}_{ir}$, measures the change in average drought conditions between the same two periods. It is constructed analogously as $\Delta \overline{D}_{ir} = \left(\frac{1}{6}\sum_{t=2015}^{2020} D_{irt}\right) - \left(\frac{1}{6}\sum_{t=2000}^{2005} D_{irt}\right)$.

In this specification, the coefficient of interest, β^{LD}, captures the long-run land-use response to changes in drought conditions. The model includes regional-level fixed effects, δ_r, which absorb region-level factors that are common to communes within the same region and may affect changes in land use between the starting and ending periods. $\Delta \varepsilon_{ir}$ is the differenced error term. All regressions are weighted using each commune's total agriculture-related area in the baseline year 2000, calculated as the sum of the areas of rice paddies, other cropland, and aquaculture. The regression is estimated at the hectares level[11], and we interpret the coefficient β^{LD} by comparing it with the mean of the land use area at baseline year (2000) to express the drought effect with 1 standard deviation increase of PDSI on land use change in percentage terms.

To quntify the extent of long-run impacts, we compare the estimates from the long-difference approach with the panel fixed-effect model. Following Schlenker and Roberts (2009) and Deschênes and Greenstone (2007), we estimate a panel fixed-effect model

[11] At baseline, 127, and 3,134 communes report zero rice and aquaculture area, respectively. The mass at zero is therefore substantial for some outcomes, particularly aquaculture, making conventional log transformations problematic. Following Chen and Roth (2023); Mullahy and Norton (2024), we estimate these outcomes in levels and scale the estimated effects by the mean of the dependent variable for interpretation.

using annual observations from 2000 to 2020:

$$Y_{irt} = \beta^{FE} D_{irt} + \alpha_i + \xi_{rt} + \varepsilon_{irt} \tag{3}$$

This model identifies β^{FE} by using the within-commune variation over time, effectively differencing out the time-invariant α_i. The inclusion of ξ_{rt} ensures that our estimates are not biased by unobserved shocks that vary at the regional level, such as regional policy changes. We cluster standard errors at the commune level.

We compare the long-difference estimate, β^{LD}, with the panel fixed-effect estimate, β^{FE}, to examine how land-use responses to drought differ across time horizons. A larger absolute value of β^{LD} is consistent with greater land-use adjustment over the long run, while a smaller absolute value indicates a weaker long-run response. Since land use itself represents a potential adaptation channel, this comparison captures differences in the extent of land-use adjustment rather than the share of drought damages mitigated through adaptation.

4 Results

4.1 Main Effects

We begin with Table 2 by examining how drought exposure reshapes land allocation across rice paddy, other crops, and aquaculture. Columns (1), (3), and (5) report the short-run effects of drought exposure on land use using the panel fixed effects specification, while columns (2), (4), and (6) report the long-run effects using the long-difference specification.

In the short run, a one-standard-deviation increase in drought exposure significantly reduces rice paddy area by 10.9 hectares, or 4.6% of the 2000 average area. In contrast, land allocated to other crops and aquaculture increases by approximately 8.0 hectares, or 2.8%, and 3.8 hectares, or 25.6%, respectively. The magnitudes of these changes suggest that land shifted out of rice production is largely reallocated toward other crops and aquaculture in the short run. In the long run, the response is concentrated in crop reallocation. Rice paddy area declines significantly by 14.4 hectares, or 3.7%, while land allocated to other crops increases significantly by 29.2 hectares, or 6.2%. Thus, the panel fixed effects and long-difference estimates yield a consistent pattern. Greater drought exposure reduces land allocated to rice paddy and increases the allocation to other crops in both the short and long run, suggesting a shift toward crop diversification as an

adaptation to drought. In contrast, we find aquacultural land area also goes up in the long run by 6.1 hectares, equivalent to 24.3% but statistically insignificant, potentially because the average effect masks substantial heterogeneity across areas with differing capacities to adapt.

4.2 Heterogeneous Effects

We next examine whether pre-existing local conditions constrain or facilitate land use adaptation to drought, focusing on surface water conditions. Tables 3 present differences in land use responses across communes along these dimensions. Surface water is the predominant source of irrigation water in Vietnam, supplying 99% of the area equipped for irrigation in 2005 (Frenken, 2012). This motivates our examination of baseline surface water availability and quality as potential moderators of land use responses to drought. We measure surface water availability using the baseline share of commune area covered by surface water and split the sample into two groups. Communes with above median shares are classified as having high surface water availability. The other half of the communes are defined as having low water availability. We extend the long difference specification by interacting drought exposure with this baseline group indicator.

Surface Water Availabillity. Panel A of Table 3 shows that drought in low water share communes is associated with a decline in rice paddy area and increases in other cropland and aquaculture area. High water share communes experience smaller declines in rice area and smaller increases in other cropland, although these differences are not statistically significant. In contrast, the interaction for aquaculture is negative and statistically significant. This indicates that aquaculture area declines in high water share communes. Figure A.1 and Figure A.2 help explain this pattern. Areas with high surface water shares are concentrated in the Red River Delta, the Mekong River Delta, and along the coast. These areas also tend to have higher water salinity. Thus, greater surface water availability does not necessarily imply better conditions for aquaculture expansion because water quality also matters.

Surface Water Quality. Panel B of Table 3 examines heterogeneity by baseline water salinity. The decline in rice area becomes larger at medium and high salinity levels, with the stronger decline at high salinity consistent with Nguyen et al. (2026). The response of aquaculture is nonmonotonic. Aquaculture area declines at low and high salinity levels but increases at medium salinity. This pattern may reflect the importance of both freshwater and brackish water aquaculture in Vietnam, where brackish water aquacul-

ture represents a substantial share of the sector.[12] Moderate salinity may therefore favor aquaculture expansion following drought, while high salinity constrains both rice and aquaculture and shifts land toward other crops.

4.3 Robustness Checks

We conduct a series of robustness checks to address potential concerns about confounding, omitted long-run factors, the choice of comparison windows, outliers, and the timing of drought exposure. Our main findings remain robust across these alternative specifications and measures.

Adding Controls. Since the long-difference specification exploits differential changes in drought exposure across communes, a potential concern is that pre-existing local characteristics correlated with drought changes may also affect subsequent land-use changes. We examine this concern by additionally controlling for baseline population density, distance to the coast, and baseline nighttime light radiance, which capture differences in settlement intensity, market access, and initial local economic activity, respectively. Columns (2), (5), and (8) of Table 3 show that the estimates remain similar after including these controls, suggesting that the main results are unlikely to be driven by these observable baseline differences.

Alternative Specification. To assess whether the baseline estimates are sensitive to short-run fluctuations in drought and agricultural land use, we estimate a panel analogue of the long-difference approach, drawing on Burke and Emerick (2016) and Cui and Zhong (2024). We average annual drought exposure and land-use outcomes over four multi-year periods: 2000–2004, 2006–2009, 2010–2014, and 2015–2019. Using these period-level averages, we estimate:

$$\overline{Y}_{i\tau} = \beta^{LD}\overline{D}_{i\tau} + \alpha_i + \delta_{r\tau} + \varepsilon_{i\tau}, \tag{4}$$

where bars denote period averages, i indexes communes, and $\tau \in \{2000, 2005, 2010, 2015\}$ identifies each period by its starting year. We estimate Equation (4) separately for rice paddy area, other cropland area, and aquaculture area. Commune fixed effects, α_i, absorb time-invariant location characteristics, while region-by-period fixed effects, $\delta_{r\tau}$, absorb changes common to communes within the same region and period. Identification therefore comes from within-commune changes in average drought exposure relative

[12] Brackish-water aquaculture accounted for an estimated 59% of Vietnam's total aquaculture production value in 2004. See FAO (2006), *Review of the Current State of World Aquaculture Insurance*, FAO Fisheries Technical Paper No. 493, Section 4.2.4, available at: https://www.fao.org/4/a0583e/a0583e07.htm.

to contemporaneous changes elsewhere in the same region. By focusing on variation across multi-year periods, the specification reduces the influence of short-run fluctuations in drought and land use.[13]

Columns (3), (6), and (9) of Table 4 show that the panel long-difference estimates retain the same signs and pattern of statistical significance as the conventional long-difference estimates across all three outcomes. A one-standard-deviation increase in drought is associated with approximately a 3.3% reduction in rice paddy area and a 5.7% increase in other cropland area, relative to their respective baseline means. Both magnitudes are close to the corresponding estimates in columns (1)–(2) and (4)–(5). The aquaculture estimate remains positive and statistically insignificant, consistent with the lack of a precisely estimated effect across specifications. While the point estimate is smaller in the panel specification, the overall inference for aquaculture remains unchanged.

Alternative Time Windows. Since long-difference estimates may be sensitive to the choice of comparison periods, we vary the baseline windows of 2000–2005 and 2015–2020. We first exclude the earliest and latest years, comparing 2001–2005 with 2015–2019. We then shift the initial window to 2001–2006 or 2002–2007 while holding 2015–2020 fixed, and shift the final window to 2014–2019 or 2013–2018 while holding 2000–2005 fixed. Figure A3 shows that the main findings in Table 2 persist: droughts reduce rice paddy area and increase other crop area, with both effects remaining statistically significant. Aquaculture estimates vary in magnitude and sign but remain statistically insignificant throughout.

Outliers. We assess sensitivity to outliers by winsorizing land-use changes at the 0.5th/99.5th, 2.5th/97.5th, and 5th/95th percentiles. Table A3 shows that the coefficient signs remain unchanged, although the estimates generally attenuate as winsorization becomes more restrictive. Importantly, the aquaculture effect remains positive and statistically significant. Given the prevalence of zero aquaculture area in our sample, however, large positive changes may capture genuine transitions from zero to positive aquaculture rather than outliers. Winsorization may therefore compress these transitions and mask part of the structural transformation toward aquaculture.

Rice Cropping Months. Since the timing of drought exposure within the crop production cycle may affect agricultural responses (Kawasaki, 2023; Sumner et al., 2026), our annual PDSI measure may mask variation in drought conditions during periods most relevant for rice cultivation. We examine this concern by using RiceAtlas to identify local rice-growing months and reconstructing PDSI exposure using only these months. We as-

[13]Regressions are weighted by baseline agricultural area, with standard errors clustered at the commune level.

sign each commune the rice calendar of the RiceAtlas polygon containing its geographic reference point. For up to three reported rice seasons, we define growing months from planting through harvest and average monthly PDSI over the union of these months within each year, counting overlapping months only once.[14] Table A2 shows that the results are very similar to those using the standard PDSI measure: drought continues to reduce rice paddy area and increase other crop area, while the aquaculture effect remains positive but statistically insignificant.

5 Land Use Adaptation Mechanism

We hypothesize that Vietnam's shift from rice toward other crops and aquaculture is driven by declining returns to rice, with land reallocation shaped by differences in input requirements and local access to water and electricity. Rice is relatively freshwater-intensive, whereas intensive brackish aquaculture relies more heavily on electricity-powered machinery. We therefore test three mechanisms. First, we examine whether drought persistently depresses rice yields. Second, we study the role of water in the decision to shift away from rice, determined by the demand for water through machinery use and supply of water through irrigation dams. Last but not least, we shed light on the role of electricity in expanding aquaculture through the demand and access of electricity.

Declining Rice Yields. One mechanism through which drought may induce land-use adaptation is by reducing rice productivity. Evidence from Vietnam suggests that drought lowers rice yields (Wang et al., 2025), which may reduce farmers' incentives to maintain land in rice cultivation and encourage reallocation toward other crops or aquaculture. We examine this mechanism using the two rice-yield measures described in Section 2. Columns (1) and (2) of Table A5 show that greater long-run drought exposure is associated with lower rice yields across both measures. A one-standard-deviation increase in drought exposure reduces rice yields by approximately 0.5%–1.1%. These results support the hypothesis that drought reduces the productivity of rice cultivation, creating an economic incentive to reallocate land toward alternative agricultural uses.

Demand and Supply of Irrigation Water. We first examine irrigation-water demand proxied by the average number of water pumps per household in each commune from the 2000 Agricultural Census, splitting communes into high- and low-use groups at the median. Panel A of Table 5 shows that the rice-area response is significantly more negative in high-use communes, where a one-standard-deviation increase in drought

[14] Figure A.4 shows the resulting cropping calendar.

exposure is associated with an estimated decline of 4.1% of the baseline mean. The point estimates indicate that other cropland expands in both groups, but the expansion is significantly smaller in high-use communes. The aquaculture response does not differ significantly between the two groups.

We next examine irrigation-water supply using baseline irrigation dam numbers. Because dams involve large fixed costs and long construction periods, baseline numbers capture irrigation infrastructure in place prior to subsequent drought exposure. We distinguish between dams in a commune's own district and dams in nearby upstream districts, using the measures described in Section 2. We then estimate separate long-difference specifications interacting the change in drought exposure with indicators for high own-district and high nearby-upstream dam numbers.

Panel B of Table 5 shows that communes with more irrigation dams in their own districts experience a smaller drought-related contraction in rice area. For a one-standard-deviation increase in drought exposure, the estimated decline is 4.0% of the baseline mean in the low-dam group, compared with 1.1% in the high-dam group, and the difference is statistically significant. Higher irrigation dam stocks in both own districts and nearby upstream districts are also associated with significantly greater expansion of other cropland. However, nearby-upstream dams do not significantly moderate the rice response, and neither dam measure significantly moderates the aquaculture response. We distinguish irrigation from hydropower dams because agricultural water supply and electricity generation involve different reservoir-management objectives and potentially competing release schedules (Olmstead and Sigman, 2015, 2026). The comparison by dam type in Appendix Table A4 finds no statistically detectable moderation by own-district or nearby-upstream hydropower dams, although the small number of hydropower-only dams in the source data warrants caution in interpreting this result.

The demand and supply results indicate that greater irrigation dependence is associated with larger contractions in rice and weaker expansion of other crops, while irrigation infrastructure provides an adaptation margin by buffering local rice losses and supporting other-crop expansion through local and upstream water supply.

Demand and Supply of Electricity for Machinery. We first examine electricity demand using baseline generator use as a proxy for reliance on supplementary power, distinguishing communes with high and low use.[15] Greater generator use may reflect electricity demand not fully met by the grid and thus identify communes where

[15]Evidence from India shows that generators buffer electricity shortages but raise electricity costs during outages (Allcott et al., 2016). Self-generation can therefore compensate for inadequate grid supply at a higher operating cost.

electricity-intensive activities are more costly to sustain. Panel A of Table 6 shows that drought reduces aquaculture significantly more in high-use communes. A one-standard-deviation increase in drought exposure significantly increases aquaculture expansion by 53.1% of the baseline mean in low-use communes but decreases it by 29.3% in high-use communes. In contrast, the rice-area and other-cropland responses do not differ significantly across groups. In high-use communes, rice area declines by 1.0% and other cropland expands by 3.1%.

We next examine electricity supply using each commune's distance to the nearest transmission line as a proxy for baseline grid accessibility. Proximity to existing grid infrastructure can reduce the cost of extending electricity supply (Dinkelman, 2011). We therefore interpret greater distance from the transmission network as weaker grid accessibility, which may constrain electricity-intensive production when drought tightens electricity supply. Panel B of Table 6 shows that drought leads to a larger decline in rice area and greater aquaculture expansion in communes closer to transmission lines. A one-standard-deviation increase in drought exposure significantly decreases rice area by 6.2% near transmission lines but increases it by 1.70% farther away. It also significantly increases aquaculture area by 62.3% near transmission lines but decreases it by 63.7% farther away. The effect on other cropland does not significantly differ by proximity to transmission lines. The electricity results are consistent with electricity constraints limiting adjustment toward aquaculture. Greater reliance on supplementary generation is associated with a weaker aquaculture response, while better grid access is associated with greater aquaculture expansion and a larger decline in rice area. This pattern suggests that electricity access facilitates reallocation from rice toward aquaculture during drought.

The yield, water, and electricity results suggest that drought-induced land reallocation reflects both lower returns to rice and the feasibility of alternative activities. Lower rice yields reduce incentives to maintain rice cultivation, while irrigation dependence increases exposure to water scarcity. Irrigation infrastructure can buffer rice losses and support expansion into other crops, while electricity access can facilitate adjustment toward aquaculture. Infrastructure can therefore shape both the ability to sustain existing production and the scope for adaptation toward alternative land uses.

6 Intersectoral Adaptation Evidences

Agriculture and non-agriculture both rely on electricity, creating an important intersectoral channel during drought. As hydropower generation falls, tighter electricity supply

can affect both sectors simultaneously, generating broader general equilibrium effects on production and labor demand.

To examine these intersectoral effects, we first use district level Enterprise Survey data from 2006 to 2016 to test whether the nonagricultural sector is also affected by drought, using manufacturing as a representative sector.[16] We estimate the effects of drought on manufacturing firms and employment and separate manufacturing into electricity intensive and less electricity intensive industries.[17] Columns (1) and (2) of Table 7 show that drought substantially reduces manufacturing activity. A one standard deviation increase in drought intensity lowers the number of manufacturing firms by 4.6 in the long run, equivalent to 20.1% of the 2006 district average. Manufacturing employment falls by 1,645 workers, or 25.9% of the baseline mean. Columns (3) through (6) show that these effects are concentrated in electricity intensive industries. The number of firms falls by 33.1% and employment falls by 37.1% in electricity intensive industries, compared with declines of 7.6% and 10.2% in less electricity intensive industries. The effects among less electricity intensive industries are therefore only about one quarter as large. This contrast is consistent with drought affecting manufacturing partly through disruptions to electricity supply.

We next examine whether competition for electricity between agriculture and manufacturing shapes machinery adaptation. Table 8 shows that drought increases the use of rice threshing machines in districts with both high and low presence of electricity intensive firms. In contrast, the expansion of aerator machines is concentrated in districts with a low presence of electricity intensive firms. This asymmetry suggests that competition from electricity intensive manufacturing can constrain electricity dependent agricultural adaptation, with aerator use expanding primarily where competing industrial demand is weaker.

We then examine whether the same intersectoral competition shapes labor adaptation. Table 9 shows that the contraction in nonagricultural employment is driven primarily by manufacturing. Nonagricultural labor declines by 18.4% in districts with a high presence of electricity intensive firms, more than three times the 5.8% decline in districts

[16] Manufacturing provides a useful setting because electricity supply disruptions may reduce firm activity, particularly in industries with greater electricity demand, and thereby reshape labor reallocation across sectors.

[17] The split is based on industry level electricity sales from 2016 to 2019 reported in *Vietnam Energy Statistics 2020*, which compiles data from Vietnam Electricity and its National Load Dispatch Center https://vepg.vn/wp-content/uploads/2022/09/English-Energy-statistics-2020-Dec-2021.pdf. We match the reported subsectors to the two digit industries in the Enterprise Survey, rank industries by electricity sales, and split the sample at the median. The electricity intensive group consists of the 13 industries above the median.

with a low presence of such firms. Agricultural labor, by contrast, increases by 2.0% and 6.4% in these two groups, respectively. Within agriculture, the increase is concentrated in crop cultivation, where labor rises by 2.5% in districts with a high presence of electricity intensive firms and by 7.4% in districts with a low presence. Aquaculture labor declines in both groups, with no statistically significant difference between them. These results suggest that drought induced contractions in manufacturing push workers out of nonagricultural employment, while the extent to which they are absorbed into agriculture depends on the capacity of the local agricultural sector. The increase in crop cultivation labor is larger in districts with a lower presence of electricity intensive firms, which are more rural and have larger agricultural sectors. These areas may therefore have greater capacity to absorb displaced workers, including workers moving from other areas within the region.

These results show that when electricity supply is vulnerable to drought, adaptation extends beyond changes within agriculture. Competition for electricity across sectors shapes the reallocation of both machinery and labor, so the capacity of agriculture to adapt depends in part on how drought affects other sectors of the local economy.

7 Conclusion Remarks

Our study provides nationally representative evidence on how Vietnam's agricultural landscape adapts to drought. In the long run, greater drought exposure reduces rice paddy area while expanding other cropland. Aquaculture also expands, particularly in areas with medium water salinity, reflecting the distinctive importance of brackish water aquaculture in Vietnam. We find that these land use responses are partly driven by declines in rice yields. In areas with high freshwater demand, especially rice producing regions, local irrigation water supplies are insufficient to fully buffer the effects of drought. For aquaculture, which relies more heavily on electricity intensive machinery, proximity to transmission lines and greater electricity availability can facilitate expansion. We also examine the role of electricity in machinery and labor adjustment. Machinery and labor reallocation depends on electricity competing demand between agriculture and the nonagricultural sector.

Our findings have important implications for drought mitigation policies, particularly for Vietnam's efforts to maintain its target for rice land. Expanding irrigation infrastructure, including water storage and irrigation capacity, could help protect rice producing areas from drought induced land conversion by improving the reliability of local water supplies. Strengthening the resilience of the electricity system is also im-

portant because drought can reduce hydropower generation at the same time that agricultural adaptation increases demand for electricity. Diversifying electricity generation toward sources that are less sensitive to drought could therefore reduce the vulnerability of agricultural production to climate shocks. Expanding transmission infrastructure could further support adaptation by improving electricity access in areas suitable for aquaculture and other electricity intensive agricultural activities. More broadly, climate adaptation policy should coordinate investments in water and electricity infrastructure while accounting for competition between agriculture and other sectors. Policies that improve reliable access to power, strengthen water supplies across sectors, and support orderly labor adjustment can help Vietnam's agricultural landscape adapt to a changing climate while limiting disruptive losses in agricultural production and employment.

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Tables

Table 1: Summary Statistics

Variable	Time Period	Mean	SD	Communes
Panel A. Drought				
Drought (PDSI)	2000-2005&2015-2020	1.570	1.073	11,027
Drought (PDSI)	2000-2020	1.64	1.79	11,027
Panel B. Land use				
Rice paddy area (ha)	2000	421.61	581.61	11,027
Cropland area (ha)	2000	507.71	729.22	11,027
Aquaculture area (ha)	2000	26.79	233.31	11,027
Panel C. Water				
Permanent surface water area (km^2)	2000	0.59	2.61	11,027
Water salinity ($mg \cdot L^{-1}$)	2000	517.20	1112.51	10,953
Panel D. Infrastructure				
Irrigation dams, own districts	2000	1.73	10.31	11,047
Irrigation dams, upstream neighbor districts	2000	1.57	11.15	11,047
Hydropower dams, own districts	2000	0.01	0.09	11,047
Hydropower dams, upstream neighbor districts	2000	0.01	0.11	11,047
Distance to nearest high voltage line (km)	2005	33.45	49.07	11,047
Panel E. Machinery, Labor				
Generator	2000	17.90	59.29	9,654
Water pump	2000	388.10	526.28	9,654
Rice thresher	2000	61.61	181.18	9,654
Aquaculture aerator	2000	3.33	37.29	9,654
Aquaculture processing	2000	0.42	6.21	9,654
Firms	2006	23.58	50.50	9,322
Firms, non electricity intensive	2006	12.54	23.72	9,322
Firms, electricity intensive	2006	11.03	29.01	9,322
Workers	2006	3,789.22	12,763.48	9,322
Workers, non electricity intensive	2006	1,641.55	5,657.37	9,322
Workers, electricity intensive	2006	2,147.67	8,634.62	9,322

Notes: Drought and land use variables report statistics for both long difference study periods, 2000 to 2005 and 2015 to 2020, as well as for the panel study period, 2000 to 2020. For all other variables, the table reports summary statistics for the baseline year. All variables are measured at the commune level, except for dams, firms and labor, and electricity generation capacity. Water, dams, irrigation, generation capacity, and population variables are measured in 2000. Transmission network variables are measured in 2005, while firm and labor variables are measured in 2006. Reverse PDSI is defined such that larger values indicate drier conditions. The sample is restricted to communes with non-missing land use data. See Section 2 for detailed variable definitions and data sources.

Table 2: Effects of Droughts on Land Use Allocation

	Rice Paddy Area		Other Crop Area		Aquaculture Area	
	Panel (1)	Long-diff (2)	Panel (3)	Long-diff (4)	Panel (5)	Long-diff (6)
Drought	-10.868***	-14.379***	8.001***	29.233***	3.831***	6.051
	(1.410)	(5.392)	(2.168)	(8.354)	(0.762)	(4.470)
Baseline Dep. Var. mean (*ha*)	421.615	421.615	507.712	507.712	26.786	26.786
Effect of 1 SD drought (% of Baseline)	-4.614	-3.659	2.821	6.178	25.597	24.236
Commune FE	Y	N	Y	N	Y	N
Year FE	Y	N	Y	N	Y	N
Region-Year FE	Y	N	Y	N	Y	N
Region FE	N	Y	N	Y	N	Y
Observations	231,567	11,027	231,567	11,027	231,567	11,027
R-squared	0.945	0.020	0.911	0.135	0.945	0.106

Notes: The dependent variable baseline means refer to land use in 2000. Relative impacts are calculated as $\left[\hat{\beta} \times SD_{\text{PDSI}} / \overline{Y}_{2000}\right] \times 100$, where $\overline{Y}_{2000}$ denotes the 2000 baseline mean of the corresponding dependent variable. For the Panel FE specifications, we use $SD_{\text{PDSI}} = 1.79$, calculated using PDSI observations from 2000–2020. For the long difference specifications, we use $SD_{\text{PDSI}} = 1.07$, calculated using PDSI observations from 2000–2005 and 2015–2020, which correspond to the periods used in the long difference analysis. Standard errors are clustered at the commune level and reported in parentheses. Statistical significance is denoted by * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$.

Table 3: Heterogeneity: Effects of Droghts by Natural Water Endowments

	Rice Paddy Area (1)	Other Crops Area (2)	Aquaculture Area (3)
Panel A: Basline Surface Water Availability			
Drought (Low Water Share)	-19.544**	52.465***	31.949***
	(8.291)	(15.916)	(5.208)
Drought × High Water Share	11.879	-30.989	-49.888***
	(13.873)	(20.896)	(10.950)
Panel B: Baseline Surface Water Salinity			
Drought (Low Salinity)	8.260	0.290	-7.075**
	(6.083)	(14.228)	(3.406)
Drought × Medium Water Salinity	-31.095**	4.358	31.756***
	(14.121)	(15.940)	(5.868)
Drought × High Water Salinity	-70.010***	117.837***	-34.111*
	(19.554)	(23.715)	(20.363)
Observations (Panel A)	11,027	11,027	11,027
R-squared (Panel A)	0.023	0.139	0.129
Observations (Panel B)	10,953	10,953	10,953
R-squared (Panel B)	0.034	0.141	0.113
Fixed effects	Region	Region	Region

Notes. This table reports heterogeneous effects of droughts by baseline natural water endowments. Panel A splits communes into low and high surface water share groups at the median. Panel B splits communes into low, medium, and high surface water salinity terciles. The omitted groups are low water share in Panel A and low salinity in Panel B. The dependent variables are rice paddy area, other crops area, and aquaculture area in hectares. All specifications include region fixed effects and are weighted by the sum of baseline rice paddy, other cropland, and aquaculture area. Communes with a zero baseline land base are excluded. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 4: Robustness: Effects of Droughts on Land Use Allocation

	Rice Paddy Area			Other Crop Area			Aquaculture Area		
	Long-diff Standard (1)	Long-diff With Control (2)	Long-diff Panel Version (3)	Long-diff Standard (4)	Long-diff With Control (5)	Long-diff Panel Version (6)	Long-diff Standard (7)	Long-diff With Control (8)	Long-diff Panel Version (9)
Drought	-14.379***	-14.460***	-13.108***	29.233***	26.293***	26.838***	6.051	7.229	1.062
	(5.392)	(5.399)	(2.861)	(8.354)	(8.252)	(5.219)	(4.470)	(9.345)	(1.763)
Baseline Dep. Var. mean (*ha*)	421.615	421.615	421.615	507.712	507.712	507.712	26.786	26.786	26.786
Effect of 1 SD drought (% of Baseline)	-3.660	-3.680	-3.336	6.178	5.557	5.672	24.240	28.957	4.256
Fixed Effects	Region	Region	Commune, Region-by-Year	Region	Region	Commune, Region-by-Year	Region	Region	Commune, Region-by-Year
Controls	-	Yes	-	-	Yes	-	-	Yes	-
Observations	11,027	11,016	22,054	11,027	11,016	22,054	11,027	11,016	22,054
R-squared	0.020	0.023	0.638	0.135	0.138	0.554	0.106	0.119	0.782

Notes: The dependent-variable baseline means refer to land use in 2000. The reported effect of a one-standard-deviation increase in drought is calculated as $\left[\hat{\beta} \times SD_{PDSI} / \overline{Y}_{2000}\right] \times 100$, where $\overline{Y}_{2000}$ denotes the 2000 baseline mean of the corresponding dependent variable. A PDSI standard deviation (SD_{PDSI}) of 1.073 is used for all three specifications. All regressions are weighted by baseline agricultural area. Columns (2), (5), and (8) additionally include baseline population density, average distance to the coast (km), and baseline average nighttime-light radiance. Standard errors are clustered at the commune level (`geoid`) in all columns. Standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 5: Mechanism: Effects of Drought and Role of Water

	Rice Paddy Area (1)	Other Crops Area (2)	Aquaculture Area (3)
Panel A: Water Demand - Water Pumps			
Drought (Low Water Pumps)	10.616	46.445***	-14.696
	(7.391)	(17.195)	(12.886)
Drought × High Water Pumps	-26.695**	-44.020**	21.450
	(10.440)	(19.395)	(14.881)
Effect of 1 SD increase in PDSI (%) – Low	2.702	9.816	-58.870
Effect of 1 SD increase in PDSI (%) – High	-4.092	0.513	27.055
Panel B: Water Supply - Irrigation Dams			
Num. of Own District Dams			
Drought (Low Own District Dams)	-15.597***	24.653***	6.253
	(5.899)	(8.960)	(4.893)
Drought × High Own District Dams	11.256**	42.774***	-1.083
	(5.582)	(15.204)	(3.756)
Num. of Neighboring Upstream District Dams			
Drought (Low NearUp Dams)	-14.340**	26.476***	5.977
	(5.596)	(8.630)	(4.636)
Drought × High NearUp Dams	0.594	66.922***	0.488
	(5.318)	(19.679)	(3.024)
Effect of 1 SD increase in PDSI (%) – Low Own Dams	-3.969	5.210	25.048
Effect of 1 SD increase in PDSI (%) – High Own Dams	-1.105	14.250	20.710
Effect of 1 SD increase in PDSI (%) – Low NearUp Dams	-3.649	5.595	23.943
Effect of 1 SD increase in PDSI (%) – High NearUp Dams	-3.498	19.739	25.898
Observations (Panel A)	9,654	9,654	9,654
Observations (Panel B)	11,027	11,027	11,027
Baseline Dep. Var. mean (ha)	421.615	507.712	26.786
Fixed Effects	Region	Region	Region

Notes. Panel A interacts drought with an indicator for a high number of water pumps. Panel B interacts drought with indicators for a high number of irrigation dams in the commune's own district and in neighboring upstream districts (NearUp). High and low groups are defined relative to the median. The main effects of the infrastructure indicators are included but not reported. The drought coefficient captures the effect for the low group. The effect for the high group is the sum of the drought and interaction coefficients. Percentage effects of a one standard deviation increase in PDSI are calculated as $100 \times SD_{PDSI} \times \beta_D / \bar{Y} * 0$ for the low group and $100 \times SD_{PDSI} \times (\beta_D + \beta * D \times H) / \bar{Y}_0$ for the high group. $SD *$ PDSI $= 1.073$, and $\bar{Y}_0$ is the 2000 baseline mean of the dependent variable reported in the table. Dependent variables are long differences in commune rice paddy, other cropland, and aquaculture area in hectares. All specifications include region fixed effects, are weighted by baseline agricultural area, and cluster standard errors at the commune level. Communes with zero baseline agricultural land are excluded. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 6: Mechanism: Effects of Drought and Role of Electricity

	Rice Paddy Area (1)	Other Crops Area (2)	Aquaculture Area (3)
Panel A: Electricity Demand - Generator Use			
Drought	-14.226	25.243*	13.265***
	(10.639)	(12.898)	(4.295)
Drought × Baseline Generator Use High	10.234	-10.371	-20.584**
	(14.676)	(17.603)	(9.909)
Effect of 1SD increase of PDSI (%) - Low	-3.620	5.335	53.137
Effect of 1SD increase of PDSI (%) - High	-1.016	3.143	-29.319
Panel B: Electricity Supply - Accessibility to Transmission Line			
Drought	6.637	31.038**	-15.901
	(10.643)	(14.145)	(10.733)
Drought × Close to Transmission Line	-30.971**	-14.123	31.448**
	(15.008)	(19.376)	(12.482)
Effect of 1SD increase of PDSI (%) - Low	1.689	6.560	-63.697
Effect of 1SD increase of PDSI (%) - High	-6.193	3.575	62.279
Baseline Dep. Var. mean (*ha*)	421.615	507.712	26.786
Observations (Panel A)	9,654	9,654	9,654
Observations (Panel B)	11,027	11,027	11,027
Fixed Effects	Region	Region	Region

Notes. Panel A interacts drought with an indicator for high baseline generator use. Panel B interacts drought with an indicator for proximity to a transmission line. Low and High correspond to indicator values of zero and one. The drought coefficient captures the effect for the low group. The effect for the high group is the sum of the drought and interaction coefficients. Percentage effects of a one standard deviation increase in PDSI are calculated as $100 \times \beta \times SD_{\text{PDSI}}/\bar{Y} * 0$ for the low group and $100 \times (\beta + \delta) \times SD * \text{PDSI}/\bar{Y} * 0$ for the high group, where β is the drought coefficient, δ is the interaction coefficient, $SD * \text{PDSI} = 1.073$, and $\bar{Y}_0$ is the 2000 baseline mean of the dependent variable reported in the table. The same baseline mean is used for both groups. Dependent variables are long differences in commune rice paddy, other cropland, and aquaculture area in hectares. All specifications include region fixed effects, are weighted by baseline agricultural area, and cluster standard errors at the commune level. Communes with zero baseline agricultural land are excluded. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 7: Additional Adaptation Evidence: Effects of Droughts on Manufacturing Firms and Labor

	All manufacturing		Electricity-intensive		Electricity-non-intensive	
	(1) Firms	(2) Labor	(3) Firms	(4) Labor	(5) Firms	(6) Labor
Droughts	-4.578***	-981.069***	-3.695***	-820.530***	-0.883**	-160.539**
	(1.166)	(224.661)	(0.891)	(195.211)	(0.311)	(50.823)
Effect of 1 SD drought (% of baseline)	-20.104	-25.878	-33.089	-37.115	-7.608	-10.159
Dep. var. mean in baseline	24.433	4067.925	11.982	2372.232	12.451	1695.692
PDSI std. dev.	1.073	1.073	1.073	1.073	1.073	1.073
Fixed Effects	Region	Region	Region	Region	Region	Region
Observations	687	687	687	687	687	687
R-squared	0.150	0.135	0.161	0.120	0.101	0.100

Notes: The sample is from the enterprise survey from 2006 to 2016 at district level. Baseline year is at 2006. Columns (1) and (2) report effects on all manufacturing establishments; columns (3)–(6) split establishments by whether their two-digit industry is classified as electricity-intensive. Dependent variables are the differences between 2016 and 2006. All specifications include region fixed effects and are weighted by baseline district-level agricultural area. Districts with zero baseline agricultural land are excluded. Effects are reported as the response to a one-standard-deviation increase in drought intensity, expressed as a percentage of the 2006 dependent-variable mean. Standard errors are clustered at the district level and reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 8: Additional Adaptation Evidence: Agricultural Machinery Adaptation

	(1) Rice threshing machines	(2) Aquaculture food processors	(3) Aerator machines
Panel A. High electricity-intensive firm districts			
Droughts	17.815***	0.110	-0.301
	(2.002)	(0.066)	(1.957)
Effect of 1 SD drought (% of baseline)	20.991	54.705	-7.105
Dep. var. mean at baseline	91.064	0.216	4.546
PDSI std. dev.	1.073	1.073	1.073
Observations	3,779	3,779	3,779
R-squared	0.120	0.007	0.026
Panel B. Low electricity-intensive firm districts			
Droughts	7.435***	0.136	40.954***
	(1.208)	(0.100)	(9.742)
Effect of 1 SD drought (% of baseline)	16.127	49.323	1585.714
Dep. var. mean at baseline	49.467	0.296	2.771
PDSI std. dev.	1.073	1.073	1.073
Observations	4,462	4,462	4,462
R-squared	0.036	0.008	0.063
Test of equality across panels			
Difference in effect (A − B, pp)	4.864	5.382	-1592.819
P-value of equality (A = B)	0.176	0.914	0.000

Notes: The sample is from the Agricultural Census survey from 2001 to 2016 at commune level. Baseline year at 2001. Columns (1)–(3) report quantities of rice threshing machines, aquacultural food processors, and aerators. Dependent variables are the differences between 2016 and 2001. All specifications include region fixed effects and are weighted by baseline commune-level agricultural area. Communes with zero baseline agricultural land are excluded. Effects are reported as the response to a one-standard-deviation increase in drought intensity, expressed as a percentage of the 2001 dependent-variable mean. The sample is split by the baseline share of manufacturing establishments in electricity-intensive two-digit industries at district level. Effects are reported as the response to a one-standard-deviation increase in drought intensity, expressed as a percentage of the 2001 dependent-variable mean. The difference row reports the Panel A minus Panel B gap in percentage points, and the p-value tests equality of the two panel effects. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 9: Additional Adaptation Evidence: Effects of Droughts and Intersectoral Labor Reallocation

	Non-agricultural		Agricultural		
	(1) All	(2) Manufacturing	(3) All	(4) Crop cultivation	(5) Aquaculture
Panel A. High electricity-intensive firm districts					
Droughts	-264.463***	-154.068***	87.283**	107.305***	-20.022***
	(55.518)	(31.496)	(31.553)	(32.527)	(5.819)
Effect of 1 SD drought (% of baseline)	-18.436	-30.216	1.971	2.495	-15.613
Dep. var. mean at baseline	1539.206	547.116	4752.813	4615.212	137.601
PDSI std. dev.	1.073	1.073	1.073	1.073	1.073
Observations	3,779	3,779	3,779	3,779	3,779
R-squared	0.080	0.116	0.144	0.132	0.041
Panel B. Low electricity-intensive firm districts					
Droughts	-50.271***	-21.100	305.477***	335.664***	-30.187
	(15.100)	(11.202)	(37.839)	(41.440)	(16.264)
Effect of 1 SD drought (% of baseline)	-5.837	-9.180	6.381	7.399	-12.051
Dep. var. mean at baseline	924.105	246.614	5136.815	4868.019	268.796
PDSI std. dev.	1.073	1.073	1.073	1.073	1.073
Observations	4,462	4,462	4,462	4,462	4,462
R-squared	0.037	0.049	0.350	0.297	0.007
Test of equality across panels					
Difference in effect (A − B, pp)	-12.599	-21.036	-4.410	-4.904	-3.562
P-value of equality (A = B)	0.001	0.005	0.000	0.000	0.672

Notes: The sample is from the Agricultural Census from 2001 to 2016 at commune level. Baseline year is 2001. Columns (1) and (3) report effects on total non-agricultural and total agricultural labor counts. Column (2) reports the manufacturing component of non-agricultural labor; the remaining non-agricultural sectors are reported in Appendix Table A6. Columns (4) and (5) report the crop cultivation and aquaculture components of agricultural labor, which sum to column (3). Dependent variables are the differences between 2016 and 2001. All specifications include region fixed effects and are weighted by baseline commune-level agricultural area. Communes with zero baseline agricultural land are excluded. The sample is split by the baseline share of manufacturing establishments in electricity-intensive two-digit industries at district level. Effects are reported as the response to a one-standard-deviation increase in drought intensity, expressed as a percentage of the 2001 dependent-variable mean. The difference row reports the Panel A minus Panel B gap in percentage points, and the p-value tests equality of the two panel effects. Standard errors are clustered at the district level and reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Figures

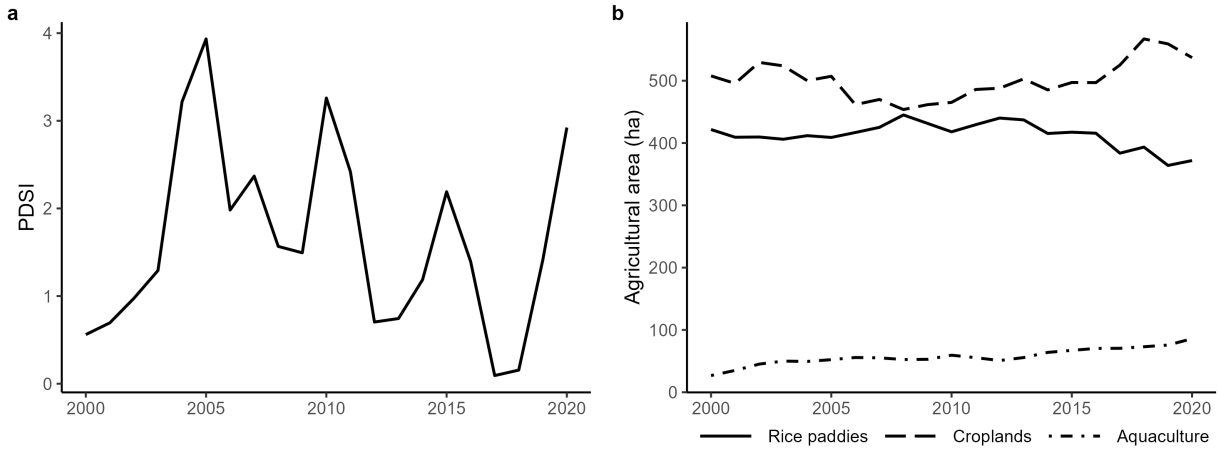


Figure 1: Drought Conditions and Agricultural Land Use in Vietnam, 2000–2020

Notes: Panel (a) plots the annual mean PDSI, with the sign reversed so that higher values indicate drier conditions, averaged across all communes in the estimation sample. Panel (b) plots the mean commune level area devoted to rice paddies, other croplands, and aquaculture, measured in hectares.

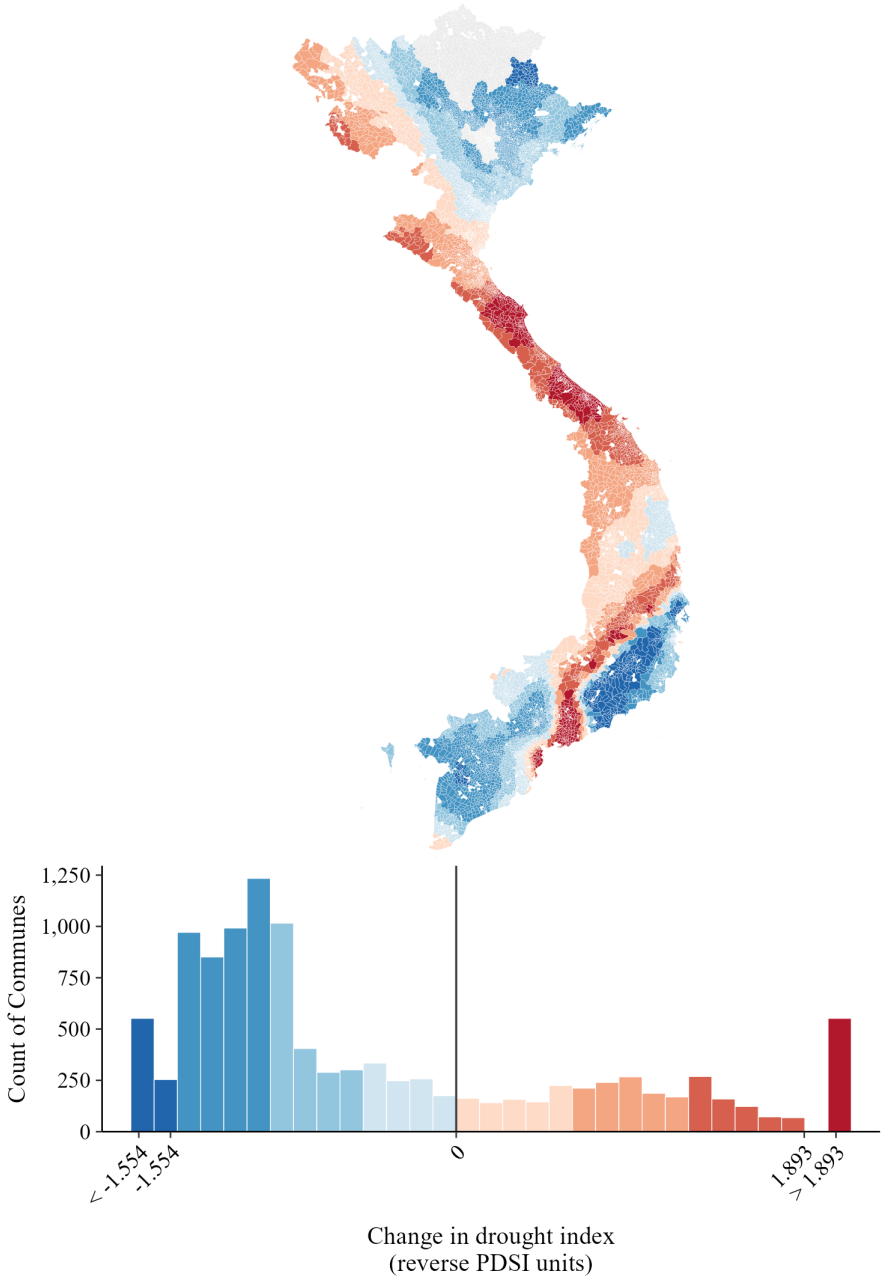


Figure 2: Long Run Changes in the Spatial Distribution of Drought Exposure

Notes: This figure presents the spatial distribution of commune-level changes in drought exposure between 2000–2005 and 2015–2020. Changes are calculated as the average over 2015–2020 minus the average over 2000–2005. Drought exposure is measured using the reverse-signed Palmer Drought Severity Index (PDSI), so that higher values indicate drier conditions. The histogram shows the distribution of these changes across communes. Changes in the bottom and top 5 percent of the distribution are pooled into the first and last bins, respectively. Communes experiencing normal drought conditions in both periods are shown in light gray.

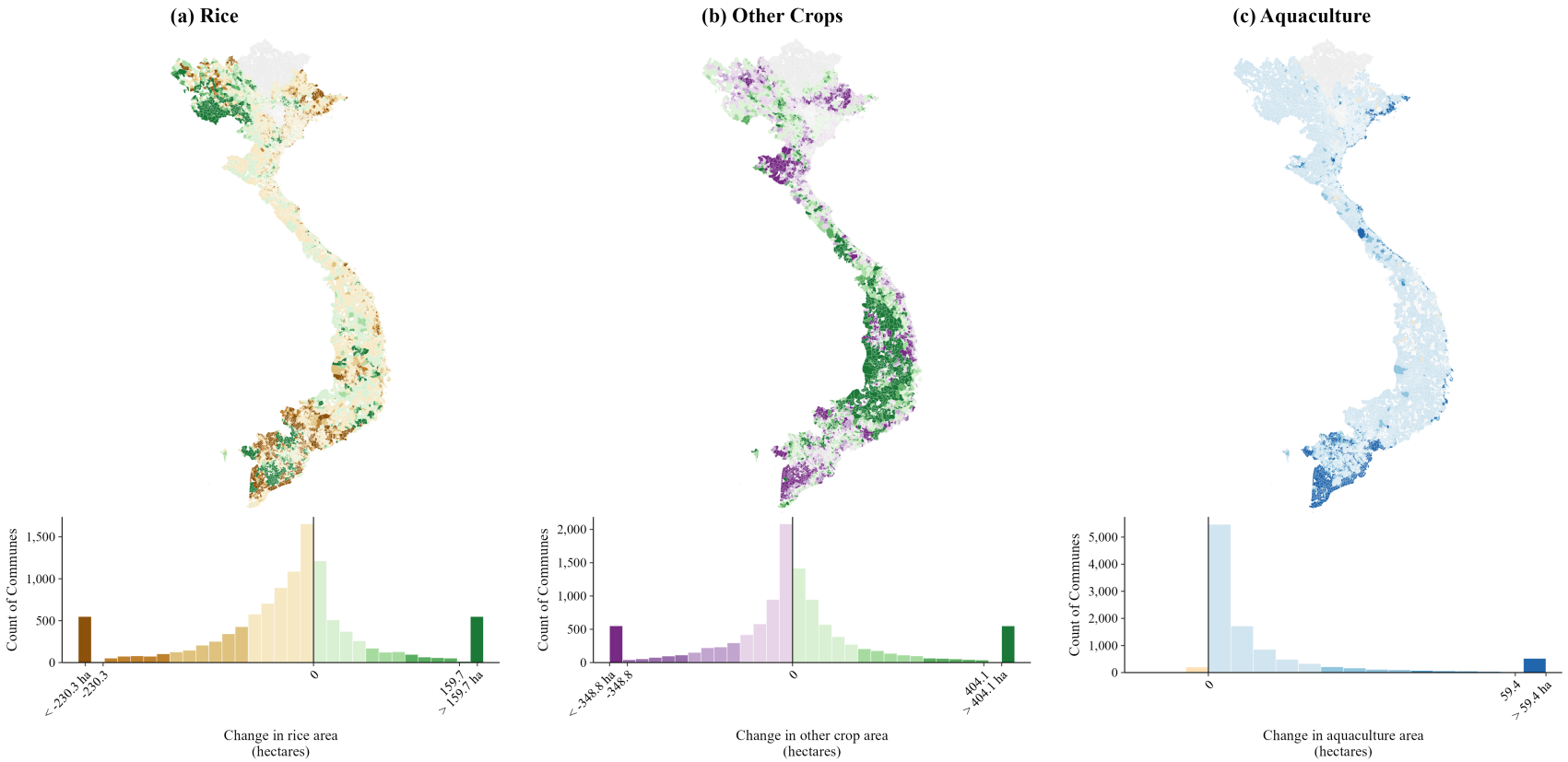


Figure 3: Long Run Land Use Changes for Rice Paddy, Other Crops, and Aquaculture

Notes: This figure presents the spatial distribution of commune-level changes in agricultural land use between 2000–2005 and 2015–2020. Changes are calculated as the average area over 2015–2020 minus the average area over 2000–2005 and are measured in hectares. Panels (a), (b), and (c) show changes in rice paddy area, other cropland area, and aquaculture area, respectively. The histograms show the distribution of these changes across communes. Changes in the bottom and top 5 percent of each distribution are pooled into the first and last bins, respectively. Communes with no land in the corresponding land use category in either period are shown in light gray.

A Appendix

A.1 Tables

Table A1: Average Commune-Level Land Use Area by Category in Vietnam, 2000–2020 (VLUCD)

Land Use Category	Average Area (km^2) Per Commnue in 2000	Average Area (km^2) Per Commnue in 2020
Residential	0.166	0.803
Rice Paddies	4.209	3.712
Croplands	5.068	5.360
Grassland	0.698	1.106
Barren	0.610	0.580
Shrub	0.823	1.072
Forest	18.465	17.002
Wetlands	0.259	0.209
Aquaculture	0.267	0.858
Water	0.753	0.617
Whole Commune Area	29.725	29.725

Notes: Values represent average land use area (in km^2) per commune. Data are derived from the VLUCD dataset and reflect commune-level averages across Vietnam.

Table A2: Robustness: Effects of Droughts on Land Use Allocation

	Rice Paddy Area		Other Crop Area		Aquaculture Area	
	Standard PDSI (1)	RiceCropMonth PDSI (2)	Standard PDSI (3)	RiceCropMonth PDSI (4)	Standard PDSI (5)	RiceCropMonth PDSI (6)
Drought	-14.379***	-12.885**	29.233***	30.876***	6.051	6.755
	(5.392)	(5.183)	(8.354)	(9.125)	(4.470)	(4.357)
Baseline Dep. Var. mean (*ha*)	421.615	421.615	507.712	507.712	26.786	26.786
Effect of 1 SD drought (% of Baseline)	-3.660	-3.279	6.178	6.525	24.240	27.059
Observations	11,027	10,996	11,027	10,996	11,027	10,996
R-squared	0.020	0.020	0.135	0.135	0.106	0.106

Notes. Columns (1), (3), and (5) use Standard PDSI. Columns (2), (4), and (6) use RiceCropMonth PDSI. Percentage effects of a one standard deviation increase in PDSI are calculated as $100 \times \hat{\beta} \times SD_{\text{PDSI}} / \bar{Y} * 0$, where $\hat{\beta}$ is the drought coefficient, $SD * \text{PDSI} = 1.073$, and $\bar{Y} * 0$ is the 2000 baseline mean of the dependent variable reported in the table. The same $SD * \text{PDSI}$ is used for both PDSI measures. Dependent variables are long differences in commune rice paddy, other cropland, and aquaculture area in hectares. All specifications include region fixed effects, are weighted by baseline agricultural area, and cluster standard errors at the commune level. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A3: Robustness: Winsorization of Land-Use Changes

	Rice Paddy Area			Other Crop Area			Aquaculture Area		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
Drought	-9.306***	-3.846*	-2.267	16.076**	7.546*	6.426**	12.300***	4.876***	1.488***
	(3.303)	(2.036)	(1.467)	(6.885)	(4.246)	(3.028)	(3.074)	(0.664)	(0.221)
Baseline Dep. Var. mean (*ha*)	421.615	421.615	421.615	507.712	507.712	507.712	26.786	26.786	26.786
Effect of 1 SD drought (% of Baseline)	-2.368	-0.979	-0.577	3.398	1.595	1.358	49.272	19.531	5.962
Observations	11,027	11,027	11,027	11,027	11,027	11,027	11,027	11,027	11,027
R-squared	0.033	0.034	0.034	0.145	0.131	0.103	0.157	0.288	0.376

Notes: Columns (1)–(3) report the same regression specification after winsorizing the dependent variable at the 0.5th/99.5th, 2.5th/97.5th, and 5th/95th percentiles, respectively. Cutoffs are calculated separately for each outcome using its unweighted distribution in the input data, before regression estimation. Winsorization replaces values outside these cutoffs with the corresponding cutoff values rather than dropping observations. The drought measure is not winsorized. The reported baseline means are mean land-use areas in 2000 and are held fixed across columns. The effect of a one-standard-deviation increase in drought is calculated as $\left[\hat{\beta} \times SD_{\text{PDSI}} / \overline{Y}_{2000}\right] \times 100$, where $\overline{Y}_{2000}$ is the corresponding baseline mean, and SD_{PDSI} equals to 1.073. All regressions are weighted by baseline agricultural area and include region fixed effects. Standard errors are clustered at the commune level and reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A4: Heterogeneous Drought Impacts by Baseline Irrigation and Hydroelectric Dams

	Rice paddy land area (1)	Other crop land area (2)	Aquaculture land area (3)
Panel A: Irrigation Dams			
Num. of Own District Dams			
Drought	-15.597***	24.653***	6.253
	(5.899)	(8.960)	(4.893)
Drought × Own District Dams	11.256**	42.774***	-1.083
	(5.582)	(15.204)	(3.756)
Num. of Near Upstream District Dams			
Drought	-14.340**	26.476***	5.977
	(5.596)	(8.630)	(4.636)
Drought × Near Upstream Dams	0.594	66.922***	0.488
	(5.318)	(19.679)	(3.024)
Panel B: Hydroelectricity Dams			
Num. of Own District Dams			
Drought	-14.426***	29.101***	6.111
	(5.438)	(8.371)	(4.516)
Drought × Own District Dams	14.877	-56.667	-6.166
	(25.269)	(112.479)	(4.205)
Num. of Near Upstream District Dams			
Drought	-13.772**	31.063***	6.464
	(5.618)	(8.546)	(4.755)
Drought × Near Upstream Dams	-9.891	-34.968	-7.302
	(16.733)	(41.324)	(5.194)
Observations	11,027	11,027	11,027
R-squared	0.021	0.140	0.106
Fixed Effects	Region	Region	Region

Notes. This table reports heterogeneous effects of droughts by baseline dam infrastructure. Panel A uses irrigation dams and Panel B uses hydroelectric dams. Within each panel, the first specification interacts drought with the number of dams in the commune's own district, while the second interacts drought with the number of dams in neighboring upstream districts. The main effect of the corresponding dam measure is included but not reported. Dependent variables are long differences in commune rice paddy, other cropland, and aquaculture area in hectares. All specifications include region fixed effects, are weighted by baseline agricultural area, and cluster standard errors at the commune level. Communes with zero baseline agricultural land are excluded. $p < 0.10$, $p < 0.05$, and $p < 0.01$.

Table A5: Effects of Droughts on Land Productivity

	Rice yield (ton/ha)	
	Administrative data (1)	Remote sensing data (2)
Drought	-0.005***	-0.010***
	(0.001)	(0.002)
Effect of 1 SD drought (%)	-0.54	-1.07
PDSI std. dev.	1.073	1.073
Observations	10,681	11,027
R-squared	0.523	0.008
Fixed Effects	Region	Region

Notes: Column (1) uses administrative rice yield data, and Column (2) uses remote sensing rice yield data. The dependent variable in both columns is the log of rice yield. Regressions are weighted by baseline agricultural area. The reported effect of a one-standard-deviation increase in drought is calculated as $100[\exp(\beta \times SD_{\text{PDSI}}) - 1]$, where β is the estimated drought coefficient and SD_{PDSI} is the standard deviation of the drought measure. Standard errors are clustered at the commune level and reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A6: Long-difference Effects of Droughts on Labor Allocation, by Sector

	Non-agricultural				Agricultural	
	(1) Manufacturing	(2) Construction	(3) Commerce	(4) Transportation	(5) Crop cultivation	(6) Aquaculture
Panel A. High electricity-intensive firm districts						
Droughts	-154.068***	-5.966	-20.995*	-15.009***	107.305***	-20.022***
	(31.496)	(4.682)	(10.179)	(3.704)	(32.527)	(5.819)
Effect of 1 SD drought (% of baseline)	-30.216	-4.562	-5.302	-19.232	2.495	-15.613
Dep. var. mean at baseline	547.116	140.331	424.917	83.745	4615.212	137.601
PDSI std. dev.	1.073	1.073	1.073	1.073	1.073	1.073
Observations	3779	3779	3779	3779	3779	3779
R-squared	0.116	0.027	0.092	0.045	0.132	0.041
Panel B. Low electricity-intensive firm districts						
Droughts	-21.100	-11.177***	-0.997	-0.705	335.664***	-30.187
	(11.202)	(2.659)	(6.326)	(1.307)	(41.440)	(16.264)
Effect of 1 SD drought (% of baseline)	-9.180	-20.128	-0.323	-1.544	7.399	-12.051
Dep. var. mean at baseline	246.614	59.585	330.794	48.977	4868.019	268.796
PDSI std. dev.	1.073	1.073	1.073	1.073	1.073	1.073
Observations	4462	4462	4462	4462	4462	4462
R-squared	0.049	0.135	0.091	0.010	0.297	0.007
Test of equality across panels						
Difference in effect (A − B, pp)	-21.036	15.566	-4.979	-17.688	-4.904	-3.562
P-value of equality (A = B)	0.005	0.013	0.115	0.000	0.000	0.672

Notes: The sample is from the Agricultural Census survey from 2001 to 2016 at commune level. Baseline year is 2001. Columns (1)–(4) report the effect on count of non-agricultral labor, breaking down to manufacturing, construction, commerce, and transportation respectively. Columns (5)–(6) report the effect on count of agricultral labor, breaking down to crop cultivation and aquaculture respectively. Dependent variables are the differences between 2016 and 2001. All specifications include region fixed effects and are weighted by baseline commune-level agricultural area. Communes with zero baseline agricultural land are excluded. Effects are reported as the response to a one-standard-deviation increase in drought intensity, expressed as a percentage of the 2001 dependent-variable mean. The sample is split by the baseline share of manufacturing establishments in electricity-intensive two-digit industries at district level. Effects are reported as the response to a one-standard-deviation increase in drought intensity, expressed as a percentage of the 2001 dependent-variable mean. The difference row reports the Panel A minus Panel B gap in percentage points, and the p-value tests equality of the two panel effects. Standard errors are clustered at the district level and reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

A.2 Figures

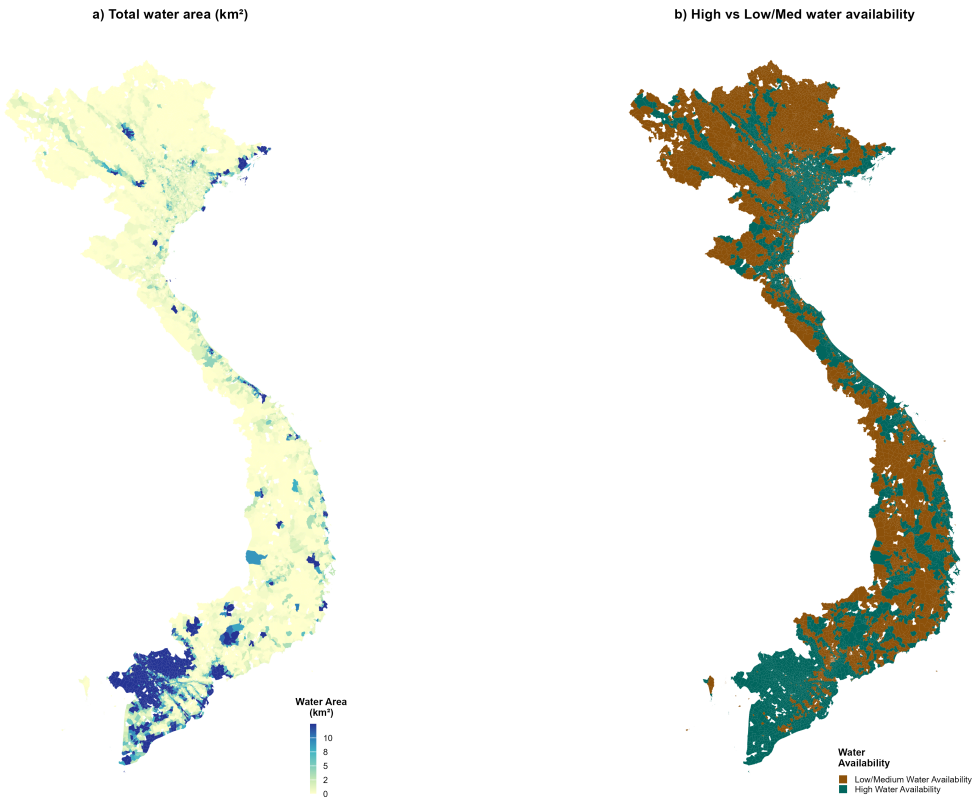


Figure A1: Spatial Patterns of Water Availability in Vietnam, 2000

Notes. Panel (a) shows total surface water area at the commune level in 2000, measured in km^2. Panel (b) classifies communes into high and low surface water availability based on the median.

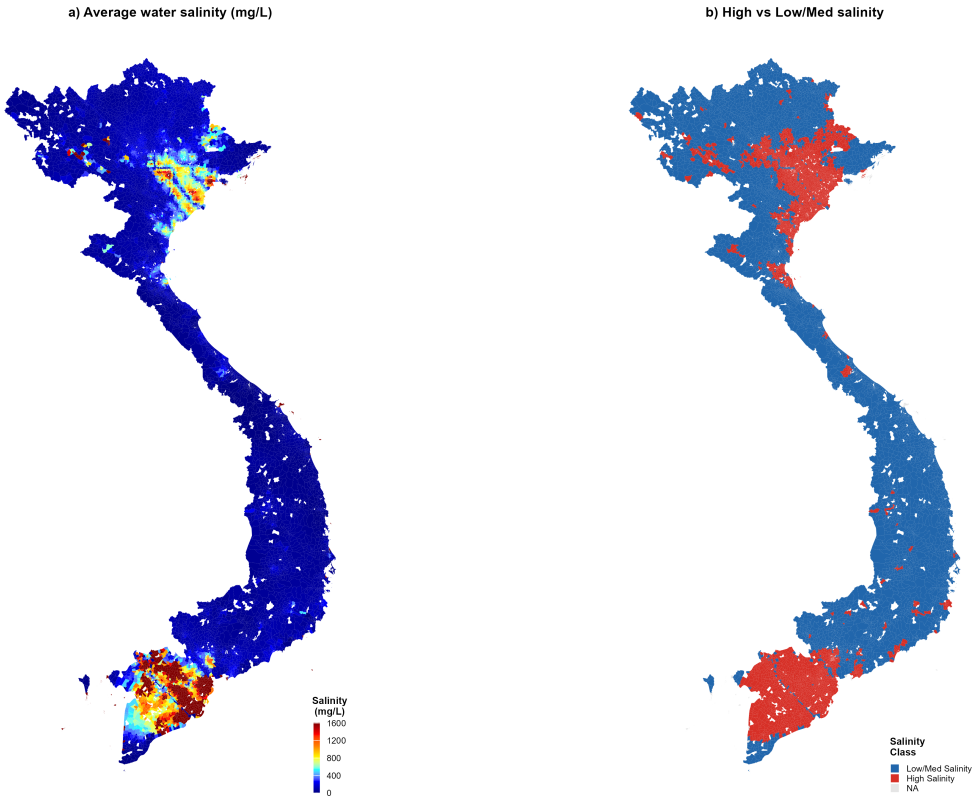


Figure A2: Spatial Patterns of Water Salinity in Vietnam, 2000

Notes: Panel (a) shows the commune-level average surface water salinity concentration (mg/L) in 2000. Panel (b) classifies communes into high-salinity versus low/medium-salinity areas.

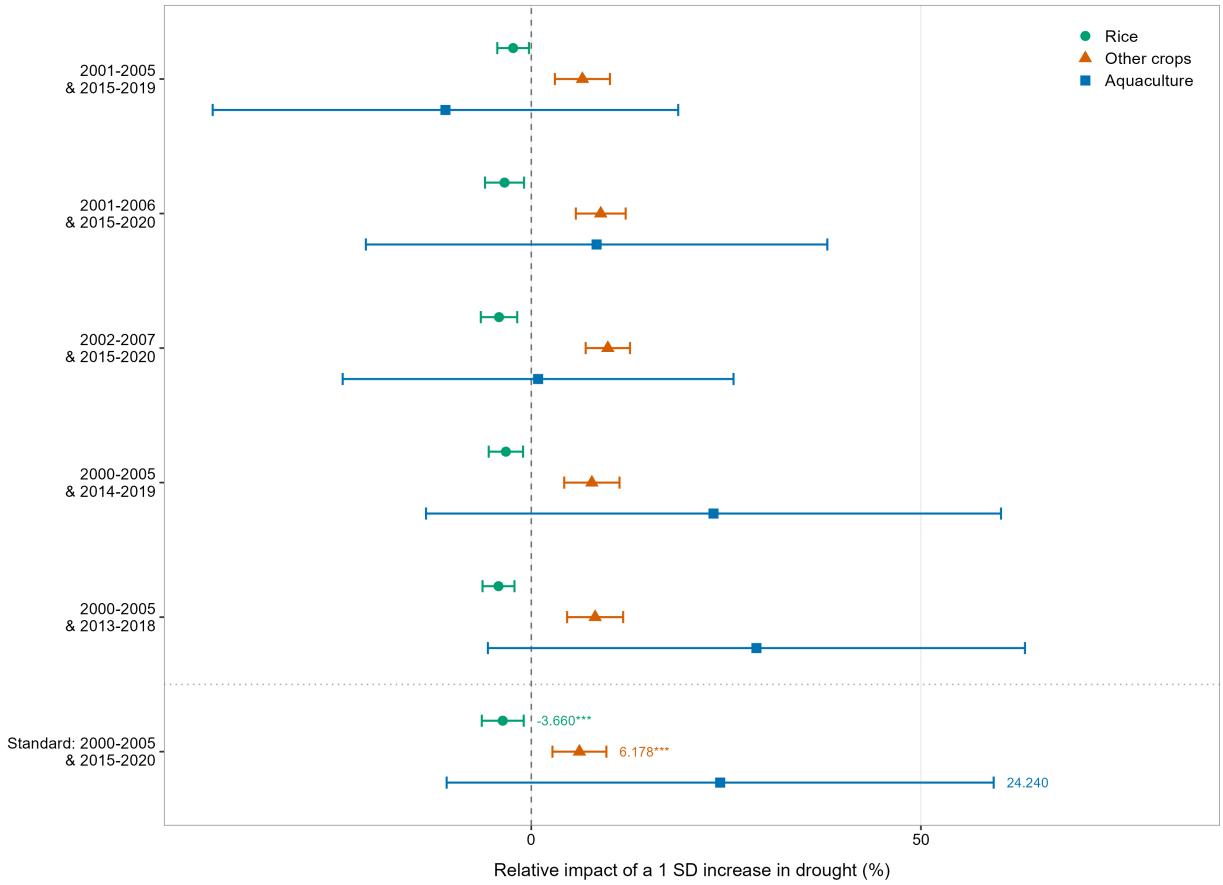


Figure A3: Robustness: Alternative Time Windows for Long-diff Specification

Notes: Markers show the estimated effects of a one-standard-deviation increase in drought, expressed as percentages of the corresponding baseline mean land-use area. Horizontal bars indicate confidence intervals. Each row uses the indicated comparison windows; the bottom row reports the baseline estimates. $^{***}p < 0.01$.

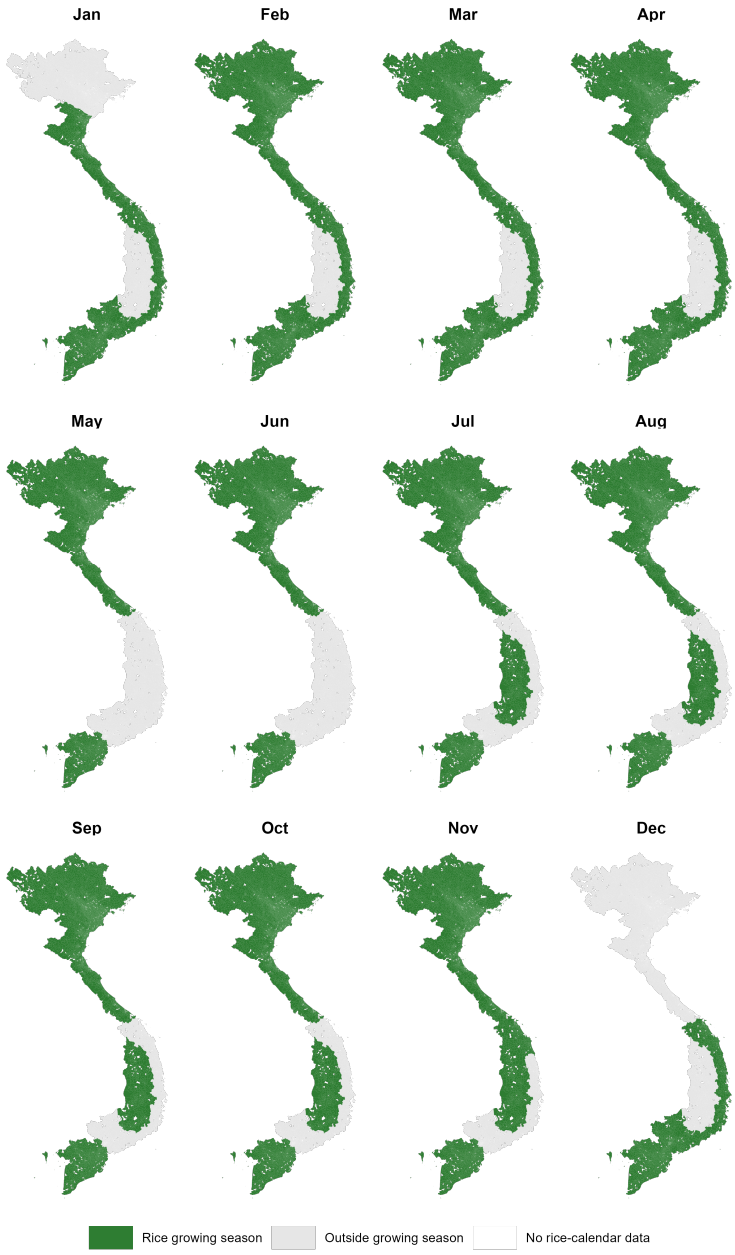


Figure A4: Rice-Growing Months in Vietnam

Notes: The figure shows commune-level rice-growing months used to construct the rice-season-specific PDSI measure. Green indicates months within the local rice-growing season, while gray indicates months outside the growing season; white denotes areas without available rice-calendar information. Growing months are defined from planting through harvest for up to three rice seasons reported in RiceAtlas. Commune-level calendars are assigned based on the RiceAtlas polygon containing each commune's geographic reference point.